

Seminar 12 - Analiză complexă

1. Să se determine mulțimea punctelor singulare și natura acestora pentru următoarele funcții:

a) $f(z) = \frac{\sin z}{z}, z \in \mathbb{C}^*;$

b) $f(z) = \frac{\sin z}{z^2}, z \in \mathbb{C}^*;$

c) $f(z) = \frac{z^5}{(1-z)^2}, z \in \mathbb{C} \setminus \{1\};$

d) $f(z) = \frac{\cos z}{z^2 - \frac{\pi^2}{4}}, z \in \mathbb{C} \setminus \left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\};$

e) $f(z) = \frac{e^{\frac{1}{z}}}{1-z}, z \in \mathbb{C} \setminus \{0, 1\};$

f) $f(z) = \sin \frac{1}{z}, z \in \mathbb{C}^*;$

g) $f(z) = z^2 \sin \frac{1}{z}, z \in \mathbb{C}^*.$

2. Să se calculeze reziduul funcției $f : \mathbb{C} \setminus \{0, 1\} \rightarrow \mathbb{C}, f(z) = \frac{e^{\frac{1}{z}}}{1-z}$ în punctul $z_0 = 0$.

3. Să se calculeze următoarele integrale complexe, folosind teorema reziduurilor:

a) $\int_{\gamma} \frac{dz}{(z^2+1)(z-1)^3},$ unde $\gamma(t) = 1+i+\sqrt{2}e^{2\pi it}, t \in [0, 1];$

b) $\int_{\gamma} \frac{e^{\frac{1}{z}}}{z} dz,$ unde $\gamma(t) = e^{2\pi it}, t \in [0, 1];$

c) $\int_{\gamma_r} \frac{e^{\frac{1}{z-a}}}{z} dz,$ unde $a \in \mathbb{C}^*, r \in (0, \infty) \setminus \{|a|\}, \gamma_r(t) = re^{2\pi it}, t \in [0, 1];$

d) $\int_{\gamma} \frac{\operatorname{tg} z}{z} dz,$ unde $\gamma(t) = \frac{1}{2} + \frac{3}{2}e^{2\pi it}, t \in [0, 1];$

e) $\int_{\gamma} \frac{\operatorname{tg} z}{z^2} dz,$ unde $\gamma(t) = 2e^{2\pi it}, t \in [0, 1];$

f) $\int_{\gamma} \frac{dz}{z^4+1},$ unde $\gamma(t) = 1+e^{2\pi it}, t \in [0, 1];$

g) $\int_{\gamma} (z-a)^{2n+1} \cos \frac{1}{z-a} dz,$ unde $\gamma(t) = 2e^{2\pi it}, t \in [0, 1], a \in \mathbb{C}, |a| \neq 2, n \in \mathbb{N};$

h) $\int_{\gamma} \frac{\sin \frac{1}{z}}{z^2(1-z)} dz,$ unde $\gamma(t) = \frac{1}{2}e^{2\pi it}, t \in [0, 1].$